



EXAMINATIONS COUNCIL OF ESWATINI
Eswatini General Certificate of Secondary Education

CANDIDATE
NAME

--

CENTRE
NUMBER

--	--	--	--	--

CANDIDATE
NUMBER

--	--	--	--

BIOLOGY

6884/02

Paper 2 Structured Questions

October/November 2025

1 hour 15 minutes

Candidates answer on the Question Paper.

Additional materials: Calculator

READ THESE INSTRUCTIONS FIRST

Write your centre number, candidate number and name in the spaces provided.

Write your answers in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do **not** use staples, paper clips, glue or correction fluid.

Do **not** write on the bar code.

Answer **all** questions.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks is 80.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
8	
Total	

This document consists of **13** printed pages and **3** blank pages.

- 1 Fig. 1.1 is a diagram of a single yeast cell in the process of reproducing.

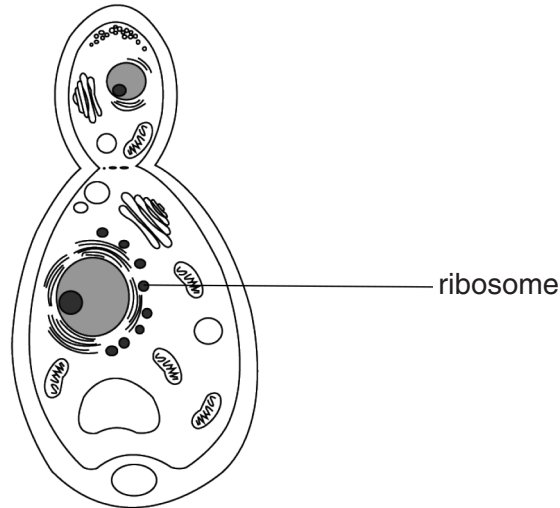


Fig. 1.1

- (a) State the type of reproduction shown in Fig. 1.1.

..... [1]

- (b) State the function of the ribosome.

..... [1]

- (c) Yeast cells respire anaerobically.

Describe how yeast respiring anaerobically is used in bread-making.

.....

 [3]

- (d) Yeast is a unicellular fungus. Many fungi are multicellular.

Describe how multicellular fungi are adapted to survive in their environment.

.....

 [3]

[Total: 8]

- 2 Fig. 2.1 shows part of the human alimentary canal and its associated organs.

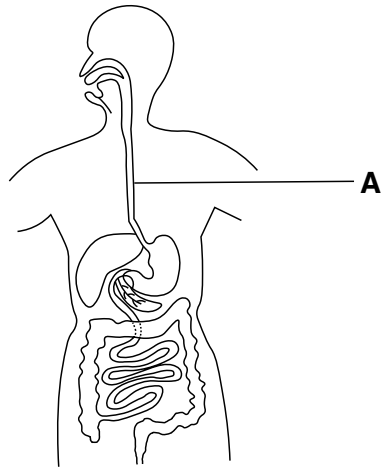


Fig. 2.1

- (a) Name the part labelled **A** in Fig. 2.1.

..... [1]

- (b) Describe the digestion of proteins in the stomach.

.....

 [3]

- (c) Excessive drinking of alcohol over a long period can lead to liver damage.

Describe the effect of liver damage on the digestion of fats.

.....

 [3]

(d) Absorption of glucose occurs in the ileum.

(i) State **two** features of the ileum that make high levels of absorption possible.

1

2

[2]

(ii) The glucose absorbed by the ileum follows the route below to reach cells in the kidneys.

ileum → liver → heart → lungs → heart → kidneys
 B **C** **D** **E** **F**

State the names of the blood vessels carrying the glucose at stages **B**, **C**, **D**, **E** and **F**.

B

C

D

E

F

[5]

[Total: 14]

[Question 3 begins on page 6]

- 3 Fig. 3.1 shows a leafy shoot. The cut end of its stem has been immersed in a red dye solution for 30 minutes.

A cross-section of the stem was then cut and observed under a microscope to see where red dye solution was present.

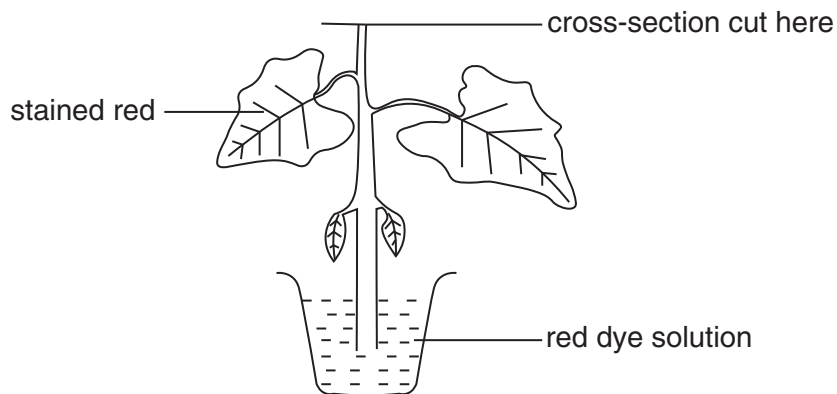


Fig. 3.1

- (a) Fig. 3.2 shows the arrangement of vascular bundles in a cross section of this stem.

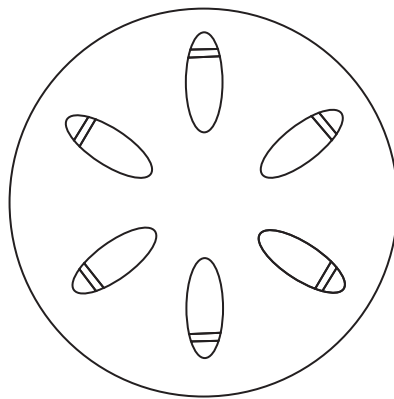


Fig. 3.2

- (i) Use a label line to indicate **one** area of the stem where red dye solution will be seen, including the name of the tissue present in that area. [1]

- (ii) After the 30 minutes, the veins in the leaves of the shoot appeared red as indicated in Fig. 3.1.

Explain how the red dye solution had reached these veins.

.....

.....

.....

.....

.....

.....

..... [4]

- (iii) After 30 minutes the cut end of the stem for the leafy shoot was taken out of the beaker of red dye solution and placed on the laboratory bench for several hours.

Describe and explain the appearance of that leafy shoot after that time.

.....

.....

..... [2]

- (b) State **two** ways in which a cross-section of the root of the plant would appear different from the cross-section of the shoot in Fig. 3.2.

1

2 [2]

[Total:9]

- 4 Fig. 4.1 shows a cross-section of human skin.

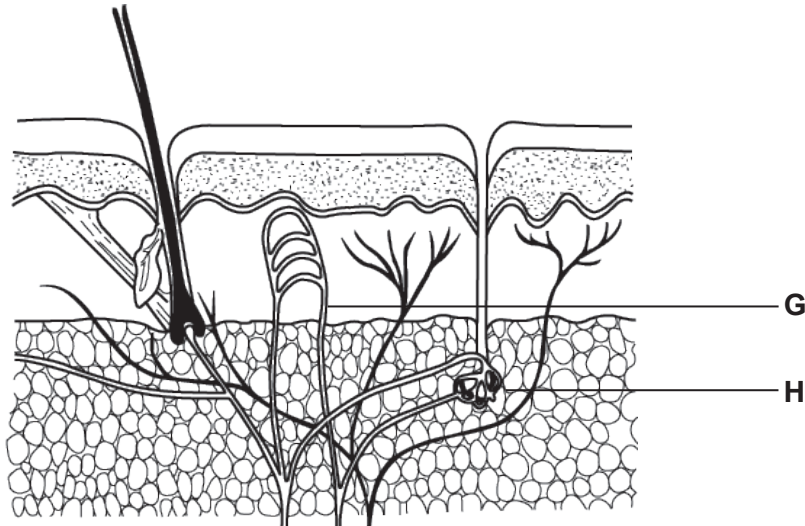


Fig. 4.1

- (a) During cold days temperatures may drop down to 5 °C in Eswatini.

Describe and explain how structure **G** and **H** in Fig. 4.1 help to keep the body temperature constant when the external temperature is 5 °C.

G

.....

.....

.....

H

.....

[5]

- (b) Sometimes a part of the body e.g. a hand may come into contact with a very hot object.

Describe the pathway in a simple reflex arc that would bring about a rapid movement of the hand away from the hot object.

.....

.....

.....

.....

.....

.....

[4]

[Total: 9]

- 5 An investigation was carried out to find the effect of light intensity on the rate of photosynthesis.

A water plant in a clear glass beaker was kept at its optimum temperature and provided with a good supply of carbon dioxide.

A lamp was placed as a source of light 10 cm away from the beaker.

The number of gas bubbles per minute given off by the water plant was counted and recorded in Table 5.1.

The investigation was repeated with the lamp at different distances from the beaker and those results were also recorded in Table 5.1.

Table 5.1

distance from lamp to beaker/cm	number of bubbles given off/minute
10	15
20	7
30	3
40	1
50	0

- (a) Suggest and explain why the water plant was kept at its optimum temperature during the investigation.

.....

 [2]

- (b) Describe and explain the effect of moving the lamp from 10 cm to 20 cm from the beaker on the rate of photosynthesis.

.....

 [4]

- (c) Some seedlings were planted out in a field with all the conditions necessary for good growth but in a soil lacking magnesium.

Explain why they failed to grow after several weeks.

.....
 [2]

[Total: 8]

- 6 Fig. 6.1 is a diagrammatic representation of some of the stages in the menstrual cycle.

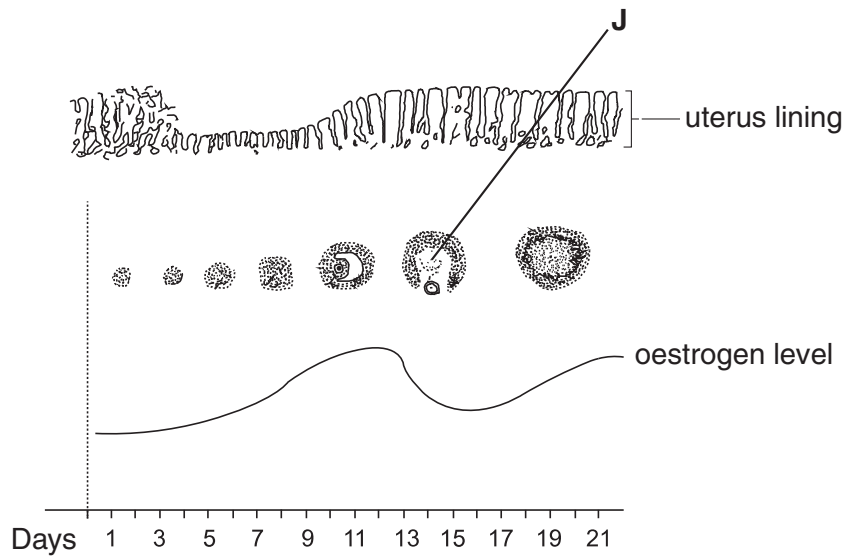


Fig. 6.1

- (a) Name the process occurring at **J** in Fig. 6.1.

..... [1]

- (b) Describe and explain what happens between days **9** and **12** to the thickness of the uterus lining.

description

.....

explanation

.....

..... [3]

- (c) Oestrogen is produced by ovaries.

State how oestrogen reaches its target organ.

..... [1]

- (d) The menstrual cycle ceases during pregnancy.

Failure to become pregnant could be due to infertility or deliberate use of a method of birth control.

- (i) Describe and explain **two** ways in which hormones in fertility drugs are used to treat infertility in females.

1

.....

.....

2

.....

.....

[4]

- (ii) State **one** mechanical method of birth control.

..... [1]

[Total: 10]

- 7 Twenty seeds were taken from the same maize plant and planted at the same time.

After the same length of time, the heights of the seedlings were measured. These are shown in Table 7.1.

Table 7.1

height range/mm	number of seedlings
0 – 19	2
20 – 39	5
40 – 59	7
60 – 79	5
80 – 99	1
100 – 119	0

- (a) State the type of variation shown by the maize seedlings.

..... [1]

- (b) State **two** factors that might cause this type of variation.

1

2 [2]

- (c) 39 maize seeds were collected from one maize cob:

28 were white and 11 were black. This showed a phenotypic ratio for seed colour of 3:1.

Using the symbols **H** to represent the allele for white seed colour and **h** to represent the allele for black seed colour, draw a genetic diagram to show how the colours of these seeds were inherited.

[5]

(d) Mutations could also lead to variation.

(i) State what a mutation is.

.....
..... [1]

(ii) State **one** factor that can increase a mutation rate.

..... [1]

(e) Describe and explain how a strain of antibiotic resistant bacteria arose from one bacterium in a population of similar bacteria that was not killed by the antibiotic.

.....
.....
.....
.....
.....
..... [3]

[Total:13]

8 Fig. 8.1 shows a simple nitrogen cycle.

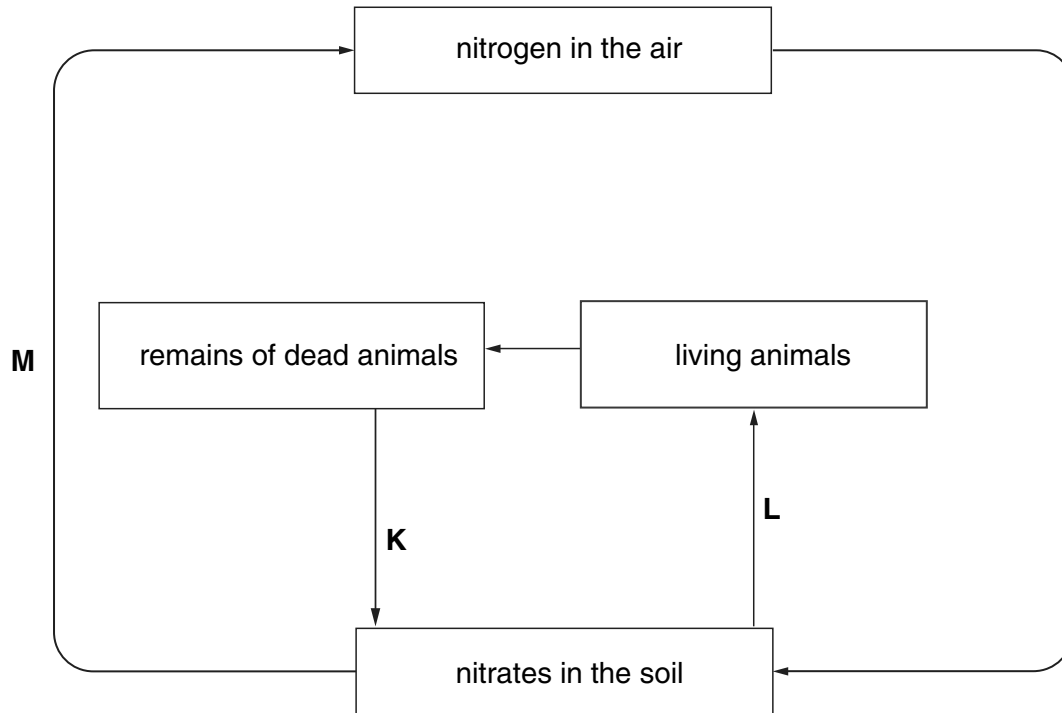


Fig. 8.1

(a) Describe what is happening at K.

.....

.....

..... [3]

(b) Name the process indicated by the line M.

..... [1]

(c) Describe how nitrates in the soil are obtained by the animal as indicated by the line L.

.....

.....

.....

..... [3]

(d) The numbers of animal pests need to be kept low. The numbers of some endangered animals need to be increased. The elephant is an endangered species.

Suggest why the extinction of elephants should be prevented.

1

2

[2]

[Total: 9]

